



AI-Powered Collections System for Geldium Finance

System Workflow Overview

AI-Powered Collections Process

Customer Data Collection

- Payment history
- Credit utilization
- Debt-to-income ratio
- Income and employment data

Risk Prediction Engine

- Logistic Regression model calculates delinquency risk score.
- Customers classified as Low, Medium, or High Risk.

Intervention Actions

- SMS reminders
- Email notifications
- Personalized repayment assistance
- Collections prioritization

Learning Loop

- Monitor intervention outcomes.
- Measure repayment improvements.
- Continuously refine recommendations.

Decision-Making and Learning Loop

End-to-End Decision Process

Inputs

- Customer financial data
- Payment behavior
- Credit usage patterns

Decision Logic

- AI model evaluates delinquency probability.

Actions

- Assign customer risk category.
- Trigger appropriate intervention strategy.

Learning

- Track response rates and repayment outcomes.
- Update business rules and model performance monitoring.

Role of Agentic AI



Autonomous Activities

Risk Scoring

Customer segmentation

Prioritizing outreach

Triggering reminders

Monitoring outcomes

Human Oversight Activities

Approval of major customer actions

Review high-impact decisions

Model governance and audits

Regulatory compliance reviews

Escalated customer cases



Responsible AI Guardrails

- Fairness Testing: Monitor outcomes across customer groups to detect bias.
- Explainability: Clearly explain why customers are classified as high risk.
- Human Oversight: Require review for high-impact decisions.
- Regulatory Compliance: Align with financial services regulations and privacy requirements.
- Data Quality Controls: Regularly validate data accuracy and completeness.
- Continuous Auditing: Periodically assess model performance and fairness.

Expected Business Impact

Business KPIs

- Reduce 30-day delinquency rate by 10%.
- Improve collections efficiency.
- Lower operational costs through automation.
- Increase accuracy of customer risk prioritization.
- Improve repayment recovery rates.

Customer Outcomes

- Earlier support for financially stressed customers.
- More personalized communication.
- Improved customer experience.
- Fairer and more consistent decision-making.
- Greater transparency in collections processes.

Expected Result

An AI-powered collections system that improves repayment outcomes while maintaining fairness, transparency, and regulatory compliance.